2.3 Exercises 10/1/26

2.3.21 $n \times m = m \times n$ (Lemma 2.3.2)

Base case either is 0, take m

$m \times 0 := 0$ by definition
$n \times m$ takes 0 and repeatedly adds to it

Inductively assume $n \times m = m \times n$. To prove $n \times m^{++} = m^{++} \times n$, with $n > 0 \wedge m^{++} > 0$ holding true

Let k be a number such that
$k^{++} = n$

$n \times m^{++} = k^{++} \times m^{++} = (k \times m^{++}) + m^{++}$

$m^{++} \times n = m^{++} \times k^{++} = (m \times k^{++}) + k^{++}$

Sub-lemma: $(a \times b^{++}) = (a \times b) + a$

Proof: base $a = 1 \rightarrow 1 \times b^{++} = b^{++} = (1 \times b) + 1$

Inductively: $a^{++} \times b^{++} = (a \times b^{++}) + b^{++}$
$= (a \times b) + a + b^{++}$
$= (a \times b) + b + a^{++}$
$= (a^{++} \times b) + a^{++}$

Using this lemma:

$(k \times m^{++}) + m^{++} = (k \times m) + k + m^{++} = (k + m) + (k + m)^{++}$

$(m \times k^{++}) + k^{++} = (m \times k) + m + k^{++} = (m \times k) + (m + k)^{++}$